EE-M-45D
August 8, 1977
SUPERSEDING
Int. Fed. Spec. EE-M-0045C(Army-GL)
March 20, 1974 and
Fed. Spec. EE-M-45B
July 8, 1971

FEDERAL SPECIFICATION

MARGARINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for margarine for use by agencies of the Federal Government.

1.2 Classification. Margarine covered by this specification shall be of the following types as specified (see 6.1):

Type I - Standard
Type II - Special
Type III - Soft
Type IV - Unsalted

The following classes are applicable to types I, II, and III:

Class 1 - All vegetable fats and oils
Class 2 - All animal fats
Class 3 - Blend of animal fats and vegetable fats and oils

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-1163 - Box, Corrugated, Fiberboard, High Compression Strength, Weather Resistant, Wax-Resin Impregnated

FSC 8945

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:43Z
Check the source to verify that this is the current version before use.

Federal Standards:

FED-STD-101 - Preservation, Packaging, and Packing Materials: Test Procedures
FED-STD-123 - Marking for Shipment (Civil agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-668 - Sanitary Standards for Food Plants

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting office.)

LAWS AND REGULATIONS

US Department of Health, Education, and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

Definitions and Standards of Identity for Margarine, Oleomargarine (21 CFR, Part 45)

(Copies may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

US Department of Agriculture

US Standards for Grades of Nonfat Dry Milk
US Standards for Grades of Dry Buttermilk
US Standards of Identity for Oleomargarine or Margarine (9CFR, Part 319)
General Specification for Dairy Plants Approved for USDA Inspection and Grading Service
Dairy Plants Surveyed and Approved for USDA Grading Service

(Copies may be obtained from the Dairy Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

United Stated Standards for Condition of Food Containers

Brucellosis Eradication - Recommended Uniform Methods and Rules, Animal and Plant Health Inspection Service, U.S. Department of Agriculture, Bulletin 91-1
Uniform Methods and Rules - Bovine Tuberculosis Eradication

(Copies may be obtained from the Dairy Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, DC 20044.)

American Oil Chemists' Society

Official and Tentative Methods

(Application for copies should be addressed to the American Oil Chemists' Society, 508 South 6th St., Champaign, IL 61820.)

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T11:43Z
Check the source to verify that this is the current version before use.

The National Research Council

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, 2101 Constitution Ave., N.W., Washington, DC 20418.)

American Public Health Association, Inc.

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1015 18th Street, N.W., Washington, DC 20036.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. Any of the ingredients permitted in the Federal Standards of Identity for Margarine, Oleomargarine or the U.S. Standards of Identity for Oleomargarine or Margarine may be used in the manufacture of margarine except the following:

(a) Edible fats or oils shall be limited to those ingredients specified in 3.1.1.
(b) Suitable edible proteins shall be limited to milk or milk products (see 3.1.2).
(c) Nutritive sweeteners shall be excluded.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26 11:43Z
Check the source to verify that this is the current version before use.

3.1.1 Fat components.

3.1.1.1 Vegetable fats and oils. Vegetable oils consisting of corn, cottonseed, peanut, sunflower, safflower or soybean or a combination of these oils shall be used. When specific vegetable oil or oils are required for dietetic use they shall be specified or when such oil is to be listed first on the ingredient label (see 6.1). All oils shall be refined, chelated and deodorized in accordance with recognized commercial practice. Any preservatives optionally permitted by the Federal Standards of Identity for Margarine, Oleomargarine and limited to the maximum amounts specified individually or in combination may be used. Oils may be partially hydrogenated. At the time of use in manufacturing the product covered by this specification, the peroxide value of all oils shall not exceed 1.5 milliequivalents per 1000 grams of oil.

3.1.1.1.1 Solid fat index. The vegetable fats or oils shall possess the following plasticity characteristics as determined by the dilatometric measurement of the oil(s) in accordance 4.4.1.

Solid fat index

TYPE I	TYPES II and IV	TYPE III	Temperature (°F)	(°C)
22.0 - 30.0	23.0 - 26.0	10.0 - 13.5	50	10
12.0 - 19.0	12.0 - 15.0	5.9 - 8.5	70	21
1.0 - 5.0	1.0 - 3.5	1.5 - 3.0	92	33

3.1.1.2 Animal fats (for class 2 and 3 only). For class 2 product, rendered animal fats or oils derived from cattle, sheep, swine or goats or any combination of two or more of such products, any or all of which may be hydrogenated, may be used. Such fats and oils shall be properly refined and deodorized in accordance with recognized commercial practice. At the time of use in manufacturing the peroxide value of each fat component shall not exceed 1.5 milliequivalents per 1000 grams of fat. Animal fats shall be prepared only in a plant which is regularly operated under the supervision of the Animal and Plant Health Inspection Service, and shall be inspected passed and marked in accordance with the Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture. For class 3 product, a blend of animal fats and vegetable fats and oils complying with 3.1.1.1 shall be used. Any preservatives optionally permitted by the Federal Standards of Identity for Margarine, Oleomargarine and limited to the maximum amounts specified individually or in combination may be used.

3.1.2 Milk and milk products. The raw milk or milk products used in the preparation of this product shall comply with all the quality requirements for No. 2 (or better) specified in the General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service. Undergrade or probational milk shall not be used. The milk shall be drawn from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the U.S. Department of Agriculture, or herds that have passed an annual tuberculosis test and meet USDA requirements for an individually certified herd, or from cows in herds located in:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

(1) A Modified Accredited Tuberculosis Area in accordance with Uniform Methods and Rules - Bovine Tuberculosis Eradication; and
(2) Either (a) A certified Brucellosis Area; or
(b) Modified Certified Brucellosis Area; or
(3) An area in the process of being certified by the USDA in accordance with USDA Animal and Plant Health Inspection Service Publication 91.1.

3.1.2.1 Pasteurized fluid milk products. Pasteurized fluid milk products such as milk, cream, skim milk and sweet cream buttermilk which meet the raw milk requirements of 3.1.2 shall at the time of use be sweet, clean and free from objectionable tastes and odors. The standard plate count shall not exceed 20,000 per ml. and the coliform count shall not exceed 10 per ml. In addition, the phosphatase activity shall be less than 1 microgram of phenol per ml. of product.

3.1.2.2 Dehydrated milk products. Nonfat dry milk and dried sweet cream buttermilk at the time of use shall meet the requirements for Extra Grade as defined in the US Standards for Grades for Nonfat Dry Milk and US Standards for Grades of Dry Buttermilk.

3.1.3 Salt. Salt shall be free flowing white refined sodium chloride and shall meet the requirements of the *Food Chemicals Codex*.

3.1.4 Water. Water shall be obtained from a supply approved as safe by State or local health authorities and shall not contain substances which adversely affect the flavor or quality of the finished product.

3.1.5 Vitamin A. Vitamin A (with or without any accompanying vitamin D) shall be in compliance with 3.4 and in addition, shall contribute no fishy or other objectionable taste or odor to the margarine.

3.1.6 Color additives. Color additives shall be any edible coloring material permitted by The Federal Food, Drug and Cosmetic Act. Provitamin A (beta carotene) shall be deemed to be a color additive.

3.2 Preparation. The product shall be made using recognized acceptable, commercial procedures from any or all of the materials in 3.1 and which will produce a product complying with the requirements of 3.3 and 3.4. Patties shall be cut 72 or 90 to the pound (158 or 198 per kg) as specified (see 6.1). The patties shall be triangular shaped or alternatively shall be square cut and labeled to comply with the requirements of 5.3.1.

3.3 Finished product.

3.3.1 Physical requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

3.3.1.1 Type I (standard). Type I standard margarine shall have a pleasing delicate taste and aroma, free from rancidity or any other foreign taste and odor. It shall have a medium yellow color and finish with a smooth body and texture, free from crumbliness, graininess, grittiness, oiliness, or droplets of moisture visible throughout the body of the product. The margarine put up in print or patty style shall retain its geometrical form and shall not become misshapen when kept exposed to ambient room temperature 70°F to 80°F (21°C to 27°C) for one hour. The finished product shall have excellent meltdown (getaway) characteristics, as evidenced by no more than a slight waxy texture in the mouth.

3.3.1.2 Type II (Special). Type II special margarine shall meet the requirements of 3.3.1.1, except that it shall possess outstanding qualities which give it an immediate meltdown (fast getaway) in the mouth with a complete absence of any discernible wax-like texture.

3.3.1.3 Type III (soft margarine). Soft margarine shall meet the requirements of type II product except that it has a distinctly softer consistency and can be spread readily on bread or crackers even at chill temperatures of 40°-45°F (4°-7°C). It shall have the physical characteristics as described and in addition shall not show any evidence of oiling off.

3.3.1.4 Type IV (unsalted margarine). Unsalted margarine shall meet the requirements of 3.3.1.1 except that it shall contain no added salt.

3.3.2 Chemical and microbiological requirements. The finished product shall meet the following chemical and microbiological requirements:

Finished product:

Fat	Not less than	80.0 percent
Salt (NaCl):		
Types I and II	Not less than Not more than	1.8 percent 2.2 percent
Type III	Not less than Not more than	1.8 percent 2.0 percent
Type IV	No added salt	
Vitamin A	Not less than	15,000 International units per pound
Vitamin D (when specified, see 6.1)	Not less than	1500 International units per pound
Milk solids not-fat	Not less than	0.50 percent
Coliform count	Not more than	10 per gram
Yeast and mold count	Not more than	20 per gram

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

3.4 Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder.

3.5 Workmanship.

3.5.1 For civil agencies. Margarine shall be prepared, processed and packed under modern sanitary conditions and by such methods as will reflect good sanitary standards of workmanship and high quality in the finished product.

3.5.2 Plant qualification (for military agencies). The product shall be processed in establishments meeting the requirements of MIL-STD-668. It shall be prepared and processed with minimum delay between the various stages in production (see 6.8). Margarine containing animal fat shall be produced in plants meeting the requirements of the Meat and Poultry Inspection Regulation of the U.S. Department of Agriculture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.

4.2.1 Plant qualification conditions. The product furnished under this document shall be unacceptable if not produced and stored in plants which currently meet the qualification condition of 3.5 (see 6.8).

4.2.2 Condition inspection of containers. When inspection of product for quality and condition is performed by the U.S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for the military, see 4.3.4.)

4.3 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated hereinafter.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:43Z
Check the source to verify that this is the current version before use.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of ingredients. Conformance of ingredients to the requirements of 3.1.1 through 3.1.6 as concerns identity (kind, grade certificates, water safety etc., as applicable) shall be determined by examination of pertinent labels, semi-annual (minimum) water approval reports from state or local health regulatory agencies, invoices or other valid documents. In addition, each of the ingredients shall be examined organoleptically, as necessary, to determine conformance to the applicable condition requirements of the above referenced paragraphs. The sample for examination shall be approximately 4 ounces (112 gram) of ingredient taken from each of five primary containers, or all containers if five or less form a lot. Nonconformance to identity, water safety, or condition requirements shall indicate an unacceptable ingredient(s) and use of such shall be cause for rejection of the involved finished product.

4.3.2 Testing of components. Testing of components described in this specification shall be as shown in table I. Requirements of test characteristics are applicable to the lot average. Nonconformance to test requirements shall be cause for rejection of the involved lot of finished product.

TABLE I. Testing of components (level B)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and rqmt. para.
Wrappers:						
Vegetable parchment paper	One 24 X 36 in. sheet (610 X 910 mm)	Pounds(kg)	S-1	Basis weight Turpentine	Nearest pound per ream (g/m^2) Nearest second	5.1.1.1(b) and 4.4.2
Wax-parchmentized paper	One 24 X 36 in. sheet (610 X 910 mm)	Pounds(kg)	S-1	Basis weight Paraffin wax Turpentine	Nearest pound per ream (g/m^2) Nearest second	5.1.1.1(c) and 4.4.2

Source: https://assist.dla.mil -- Downloaded: 20261026111:43Z
Check the source to verify that this is the current version before use.

TABLE I. Testing of components (level B) (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and rqmt. para.
Wrappers (cont'd):						
Grease-proof paper	One 24 X 36 in. sheet (610 X 910 mm)	Pounds(kg)	S-1	Basis weight	Nearest pound per ream (g/m^2)	5.1.1.1(d) and 4.4.2
				Turpentine	Nearest second	
Fiberboard box pads and liners	Two 12 X 12 in. panels (305 X 305 mm) or equivalent in total area	Box pads and liners	S-1	Wax melting point (when applicable)	Nearest degree C or F	5.2.1(a) and 4.4.2
Wax or parchmentized liner	3 sq yds (m^2)	Pounds(kg)	S-1	Wax melting point	Nearest degree F (degree C)	5.2.1(b) and 4.4.2
				Wax weight	Nearest lb/ream	
				Paper basis weight	Nearest lb/ream (gr per sq meter)	
				Waxed paper bursting strength	Nearest pound (gram)	
				Tearing resistance	Nearest gram	
Polyethylene bag	One bag	Number of bags	S-1	Thickness 2/	Nearest 0.0001 in. 0.001 mm	5.2.1(c)

Source: https://assist.dla.mil -- Downloaded: 20260-20T11:43Z
Check the source to verify that this is the current version before use.

TABLE I. Testing of components (level B) (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and rqmt. para.
Wrappers (cont'd):						
Vegetable oil or animal fat	1/	Primary containers		Peroxide 3/ value	Nearest 0.1 meq/kg of oil	4.4.1 and 3.1.1
				Solid fat index	Nearest 0.5 SFI	4.4.1 and 3.1.1.1.1
Fresh milk and milk products	1/	Primary containers		Phosphatase	Pass or fail	4.4.1 and 3.1.2.1
				Standard plate count	Pass or fail	4.4.1 and 3.1.2.1

1/ The sample for examination shall be approximately 1-pound (454 gram) composite of vegetable oil derived from five primary containers or all containers if less than five form a lot.

2/ In lieu of performing test for thickness, conformance to thickness requirement may be ascertained by examination of pertinent labels, invoices or other valid documents. If nonconformance is indicated the involved quantity of bags shall be rejected.

3/ Samples of vegetable oil sent to any government approved laboratory for peroxide value testing shall be in containers brim filled to capacity so as to minimize contact with the air and shipped with sufficient dry ice to last until arrival at the laboratory. After receipt the oil shall be held at 0°F (-18°C) or below until tested.

4.3.3 Process examination. Unless otherwise specified (see 6.1) examination shall be performed during processing to determine compliance with requirements of 3.2, as concerns preparation and formulation. A formulation or manufacturing record shall be maintained containing information pertinent to the kind of components specified and kind and amounts of optional ingredients permitted for each batch or lot or product produced and made available for review. Nonconformance to above referenced requirements, reflected either by actual examination or by examination of records, shall be cause for rejection of the involved quantity of finished product.

4.3.4 Examination of end item. Classification of defects found during examination shall be in accordance with table III through V, and 4.3.4.1 to 4.3.4.3. The examination criteria shall be as shown in table II and the acceptable quality levels (AQLs) for tables III and V shall be expressed as defects per hundred units and for table IV as percent defective.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:43Z
Check the source to verify that this is the currentversion before use.

TABLE II. End item examination criteria (level B)

Table	Inspection level	Sample unit	Lot size expressed in	AQLs	
				Major B	Minor
III	S-3	A package one or two pound (0.45 or 0.91 kg) print, a one pound (0.45 kg) package of one quarter pound (113 g) prints, two 1/2 pound (227 g) trays of patties, or five pounds (2.3 kg) of packaged patties	Pounds (kg)	1.0	4.0
IV	S-2	A packaged one or two pound (0.45 or 0.91 kg) print, a one pound (0.45 kg) package of one quarter pound (113 g) prints, two 1/2 pound (227 g) trays of patties, or five pounds (2.3 kg) of packaged patties	Pounds (kg)	-	2.5
V	S-2	A packaged one or two pound (0.45 or 0.91 kg) print, a one pound (0.45 kg) package of one quarter pound (113 g) prints, two 1/2 pound (227 g) trays of patties, or five pounds (2.3 kg) of packaged patties	Pounds (kg)	1.5	10.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE III. Examination of packaging (level B) 1/

Category	Defect	Defects applicable to wrappers and pouches	Defects applicable to cartons	Defects applicable to waxed or parchment paper separators, waxed paper packaging and chips, covers, trays or cartoning of patties only
Major B				
151	Not types, style, or size specified	X	X	X
152	Missing	X	X	X
153	Exterior not clean	X	X	X
154	Crushed or misshapen, showing exposed product surface	X	X	X
155	Not completely wrapped or inclosed	X	X	X
156	Any surface contacting product not clean	X	X	X
157	Not packaged in layers as specified	-	-	X
158	Nomenclature missing, incorrect, or illegible	X	X	X 2/
Minor				
201	Crushed or misshapen with no exposed product surface	X	X	X
202	Labeling information other than nomenclature missing, incorrect, or illegible	X	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE III. Examination of packaging (level B) 1/ (cont'd)

Category	Defect	Defects applicable to wrappers and pouches	Defects applicable to cartons	Defects applicable to waxed or parchment paper separators, waxed paper packaging and chips, covers, trays or cartoning of patties only
Minor (cont'd)				
203	Coating missing or not uniform	-	X	X
204	Patties not placed on coated side of chips 3/ (see 5.1.1.3.1)	-	-	X
205	Weight of patties per tray not as specified (see 5.1.1.3.1) 3/	-	-	X
206	More than one patty placed on a single chip 3/ (see 5.1.1.3.1)	-	-	X

1/ No more than three sample units shall be taken from one shipping container.

2/ This defect is to be scored on the individual patty only when patty is not triangular shaped (see 3.2 or 5.1.1.3.1).

3/ Examine one pound (454 g) amounts from each sample unit.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:43Z
Check the source to verify that this is the currentversion before use.

TABLE IV. Examination of net weight and patty count per pound (level B) 1/

Category	Defect
Minor	
201	Net weight more than two percent under specified weight (prints of one pound (0.45 kg) or less). Report to nearest 1/8-ounce (gram).
202	Net weight more than one percent under specified weight (prints or packages of more than one pound (0.45 kg), but not more than ten pounds) (4.5 kg). Report to nearest 1/4-ounce (gram).
203	Patty count not as specified. (Not applicable to ready-to-serve patties placed individually on chips.) 2/

1/ The lot shall be rejected if sample data indicate lot average net weight or lot average patty count, as applicable, are less than specified net weight or count, except ready-to-serve patties cut 72 or 90 to the pound (0.45 kg) and placed individually on chips, the count shall average not more than 95 nor less than 85 patties per pound (0.45 kg) for the 90 count and not more than 77 or less than 67 for the 72 count.

2/ One pound out of each sample unit shall be examined for this defect.

TABLE V. Examination for product characteristics 1/ 2/

Category		Defect
Major B	Minor	Types I, II and III
151		Not pleasing delicate taste or aroma
152		Evidence of oiliness
153		Not characteristic uniform yellow color
154		Presence of visible moisture droplets throughout body of the product
155		Evidence of crumbliness
156		Evidence of graininess
157		Evidence of grittiness
	201	Print or patty shape misshapen or does not retain geometrical form (see 3.3.1.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE V. Examination for product characteristics 1/ 2/ (cont'd)

Category		Defect
Major B	Minor	
		Type I - (standard) only
158		Does not contain excellent melt down characteristics as evidenced by no more than a slight waxy texture in the mouth 3/
		Type II (special) and type IV (unsalted) only
159		Melt down not immediate or presence of any discernible wax-like texture 3/
		Type III (soft) only
160		Not a soft consistency
161		Does not readily spread on bread or crackers (see 3.3.1.3)
162		Evidence of oiling off
		Type IV (unsalted) only
163		Contains added salt

1/ Margarine shall be tempered to an internal temperature of 70°F (21°C) prior to examination of product characteristics except for defect 201 (see 3.3.1.1).

2/ Evidence of rancidity or any other foreign taste or odor or presence of foreign material (e.g., wood, paper, rust, grease, paint, glass or metal particles, dirt, or other extraneous material) shall be the basis for rejection of the lot.

3/ When a level teaspoon of sample unit is placed in the mouth.

4.3.4.1 Examination of the filled and sealed shipping container. Examination of the shipping container shall be in accordance with the quality assurance provisions contained in the appendix to PPP-B-636 or PPP-B-1163 as applicable, for level B pack, with the following additions to the table of examination: Major; marking missing, incorrect or illegible; Minor; net weight exceeds 60 pounds (27 kg) as indicated by marked net weight. When the product is packaged as uncartoned prints or patties, the following defects shall be included in the examination, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

Major: - (1) Product not protected by one of the three specified methods (see 5.2.1, (a), (b), or (c)).
(2) Gusseted plastic bag, when applicable, not as specified or not sealed as specified.
(3) Liners or gusseted plastic bag torn.
(4) Liners or gusseted plastic bag unclean.

Minor: - (1) Top or botom pads missing or not as specified (linear measurements shall be reported to the nearest 1/32 inch) (millimeter).

When applicable, containers reinforced with other than nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape shall not be acceptable. Examination of level C packing shall be in accordance with the examination criteria of the aforementioned specifications except that only the defects pertaining to marking shall apply.

4.3.5 Sampling procedure and acceptance criteria for testing of finished product. The finished product shall be tested for the characteristics specified in 3.3.2. Procedures for testing for fat, salt, vitamin A and D (see 6.4), milk solids not fat, coliform and combined yeast and mold counts shall be in accordance with 4.4. Test requirements shall be lot average requirements. The sample for testing shall be approximately a one pound (0.45 kg) composite derived from the number of sample units (i.e.,; 1 or 2 pound prints, cartoned quarters, 5 pound packages of patties or two 1/2 pound trays of patties) (i.e.; 0.45 kg or 0.9 kg prints, cartoned quarters, 2.3 kg package of patties or two 0.45 kg trays of patties); indicated by inspection level S-2. Lot size shall be expressed in sample units. Test results shall be reported as follows: fat, MSNF, and salt to the nearest unit or decimal place as specified in the referenced test requirements paragraph; vitamin A and D to the nearest 50 USP units and all microbiological counts in accordance with Standard Methods for the Examination of Dairy Products. Nonconformance with one or more test requirements shall be cause for rejection of the lot.

4.4 Test methods. Examination and test procedures and controls which differ from those specified herein unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to the specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein shall govern.

4.4.1 Chemical and microbiological requirements. Chemical and microbiological analyses shall be made in accordance with methods from publications listed below:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

Ingredients:

Test	Source	Method
Peroxide value of oil	2/	Cd 8-53
Solid fat index	2/	Cd 10-57
Phosphatase activity of milk and skim milk products	Chapter: Phosphatase Methods to Determine Pasteurization 3/	Scharer Rapid Phosphatase Test
Standard plate count	3/	
Coliform count	Chapter: Microbiological Methods for Butter Section: Bacteriological Analyses Coliform Group 3/	Coliform Count Using Violet Red Bile Agar

Finished product:

Test	Source	Method
Fat	Chapter: Dairy Products Section: Butter 1/	Fat, indirect method
Salt	Chapter: Dairy Products Section: Butter 1/	Salt
Milk solids - not fat	Chapter: Dairy Products Section: Butter 1/	Subtract percent salt from the percent residue in the test for fat, indirect method
Vitamin A	Chapter: Vitamins and other Nutrients Section: Chemical Methods 1/	Vitamin A in Margarine
Vitamin D	Chapter: Vitamins and other Nutrients Section: Bioassay Methods 1/	Vitamin D
Coliform count	Chapter: Microbiological Methods for Butter Section: Bacteriological Analysis Coliform Group 3/	Coliform Count Using Violet Red Bile Agar

Source: https://assist.dla.mil -- Downloaded: 2025-10-29 11:43Z
Check the source to verify that this is the current version before use.

Ingredients: (cont'd)

Finished product: (cont'd)

Test	Source		Method
Yeast and mold count	Chapter:	Microbiological Methods for Butter	Yeast and Mold Counts
	Section:	Microbiological Methods 3/	

1/ Official Methods of Analysis of the Association of Official Analytical Chemists.

2/ Official and Tentative Methods of the American Oil Chemists Society.

3/ Standard Methods for the Examination of Dairy Products.

4.4.2 Paper tests. Paper shall be tested in accordance with the following methods of FED-STD-101:

Test	Method No.
Basis weight of paper and paperboard	5022
Melting point of wax	4017 (D 127-63)
Bursting strength of sheet materials	2007
Tearing resistance (Elmendorf)	2036
Oil transmission test for packaging materials (procedure A)	3017

5. PREPARATION FOR DELIVERY

5.1 Packaging. Types I, II and IV product shall be packaged in accordance with level B or C, as specified (see 6.1). Type III product shall be packaged in one pound (0.45 kg) containers in accordance with level C only. Materials used for packaging shall impart no taste or odor to the product. Materials incorporated in the wrappers shall comply in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and regulations promulgated thereunder.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

5.1.1 Level B. Margarine shall be packaged in accordance with 5.1.1.1, 5.1.1.2 or 5.1.1.3, as applicable.

5.1.1.1 Prints. As specified (see 6.1), 1/4 pound, one pound, or two pounds (113, 454 or 908 grams) prints of the product shall be completely wrapped in one of the materials specified below:

(a) A commercial foil-paper combination wrapper.
(b) Vegetable parchment having a minimum average basis weight (see 4.4.2) of 26 pounds per ream (24 by 36 - 500) (42 g per sq m) and a minimum average turpentine test of not less than 600 seconds when tested in accordance with method 3017 procedure A (see 4.4.2)..
(c) A sheet of waxed-parchmentized paper. The parchmentized paper shall have a minimum average basis weight of 26 pounds per ream (24 by 36 - 500) (42 g per sq m) and be waxed with not less than three pounds (1.4 kg) of paraffin wax per ream (see 4.4.2). The waxed parchmentized paper shall have a minimum average turpentine test of not less than 600 seconds (see 4.4.2).
(d) A sheet of greaseproof paper having a minimum average basis of 26 pounds per ream (24 by 36 - 500) (42 g per sq m) and a minimum average turpentine test of not less than 600 seconds (see 4.4.2).

5.1.1.1.1 Intermediate carton for prints. Four wrapped 1/4-pound (113 g) prints shall be packaged in a commercial, coated, intermediate carton. When specified (see 6.1), a one-pound (0.45 kg) print or one two-pound (0.91 kg) print shall be packaged in a commercial coated intermediate carton.

5.1.1.2 Pouches. One pound (0.45 kg) of the product shall be sealed in a pouch formed from one of the following materials:

(a) A vinyl film or plastic film made from a blend of polyvinyl chloride or vinyl-chloride-acetate copolymer with nitrile rubber of Buna N type rubber.
(b) A rubber hydrochloride base film.

5.1.1.3 Patties (see 6.6.6). The patties shall be packaged in layers in five pound (2.3 kg) units and placed in commercially acceptable waxed paper, or crated carton. The layers shall be separated by a sheet of waxed or parchment paper.

5.1.1.3.1 Patties on chips. Alternatively, when specified (see 6.1), patties cut 90 to the pound (0.45 kg) shall be placed on chips (see 6.6.6) made from plastic or bleached sulfite paperboard. Paperboard chips shall be coated on one side with polyethylene, and the patties shall be placed on the coated side. Patties shall be covered with a piece of waxed or parchment paper. One-half pound (0.23 kg) of patties plus or minus one ounce (28 g), shall be placed on a tray. The material for chips, covers, and trays shall be that which is normally used in good commercial practice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. Margarine shall be packaged in a manner to insure against damage and deterioration during shipment from the supply source to the first consuming activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Not more than 60 pounds (27.2 kg) of margarine shall be packed in a snug-fitting shipping container in accordance with level B or C, as specified (see 6.1).

5.2.1 Level B. The shipping container shall be a fiberboard box constructed, closed, and reinforced in accordance with style RSC-L, V3c, V3s or V4s, of PPP-B-636. Alternatively, the shipping container shall be a snug-fitting, wax-impregnated, fiberboard box constructed in accordance with style RSC, class I, type SWCFI, grade 275 of PPP-B-1163. Closure shall be made with hot melt adhesive in accordance with the appendix of PPP-B-636. The wax-impregnated box shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636. The shipping container shall be provided with top and bottom pads of the same material as the shipping container not more than 1/16-inch (1.6 mm) smaller than the inside length and width of the box. Uncartoned prints and patties shall be protected from the interior surfaces of the liner and pads as specified in (a), (b) or (c) below:

(a) The inside surfaces of the liner and pads shall be coated with a fully refined paraffin wax having a melting point of not less than 125°F (52°C) (see 4.4.2).

(b) The shipping container shall be completely lined on the inside with a one or two-piece liner of waxed paper so placed as to separate the margarine from the top, bottom, sides and ends of the container. The liner material shall be bleached or semibleached deodorized kraft or sulfite paper or parchmentized paper having a basis weight of 25 pounds per ream (24 by 36 - 500) (41 g per sq m) (see 4.4.2) before waxing. The sheet shall be waxed on both sides with not less than nine pounds (4 kg) per ream of fully refined paraffin wax having an average melting point of not less than 125°F (52°C) (see 4.4.2). The waxed paper shall have an average bursting strength of not less than 22 pounds per square inch (10 kg per 6.5 cm^2) (see 4.4.2) and an average tearing resistance of not less than 18 grams (see 4.4.2).

(c) The contents shall be inclosed in a gussetted plastic bag constructed of 0.002-inch (0.05 mm) thick polyethylene complying with L-P-378. The bag shall fit the interior of the shipping container snugly and shall be closed by heat sealing or taping.

5.2.1.1 Nonmetallic reinforcement of shipping containers. When specified (see 6.1 and 6.3), the shipping container specified in 5.2.1 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636, except that for V3c, V3s and V4s boxes two reinforcing bands may be used, one lengthwise and one girthwise.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

5.2.1.2 Alternate shipping containers. When specified (see 6.1 and 6.5), the shipping container shall be a fiberboard box, constructed, closed, and reinforced in accordance with style RSC-L, W5c or W5s of PPP-B-636. Alternatively, the shipping container shall be a snug-fitting, wax-impregnated, fiberboard box constructed in accordance with style RSC, class I, type SWCFI, grade 200 of PPP-B-1163. Closure shall be made with hot melt adhesive in accordance with the appendix of PPP-B-636. The wax-impregnated box shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636. Pads and lining shall be in accordance with 5.2.1.

5.2.2 Level C. The shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Labeling and marking.

5.3.1 Square patties. Square patties shall be labeled clearly to identify the product in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

5.3.2 Unit containers. Suitable commercial labeling in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder will be acceptable.

5.3.3 Shipping containers.

5.3.3.1 Civil agencies. Shipping containers shall be marked in accordance with FED-STD-123 and as specified in 5.3.4.

5.3.3.2 Military agencies. Shipping containers shall be marked in accordance with MIL-STD-129 and as specified in 5.3.4.

5.3.4 In addition to the markings specified in 5.3.3.1 and 5.3.3.2 as applicable, the following precautionary marking shall be printed or stenciled on top of the shipping container in bold capital letters between 1 and 1-1/2 inches high, all letters being of the same height.

PERISHABLE-KEEP

REFRIGERATED

(45°F (7°C) or lower)

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

Source: https://assist.dla.mil - Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

(a) Title, number and date of this specification.
(b) Whether type I, II or III, class 1, 2 or 3 or type IV, class 1 product is required (see 1.2).
(c) When a specific vegetable oil(s) is required for dietetic feeding (see 3.1.1).
(d) When a specific oil is to be listed first on the ingredient label (see 3.1.1).
(e) When added vitamin D is required (see 3.3.2).
(f) When in-process examination is not required by civil agencies (see 4.3.3).
(g) Patty count per pound or size of prints (see 3.2 and 5.1.1).
(h) Applicable levels of packaging and packing (see 5.1 and 5.2).
(i) When packing specified in 5.2.1.1 or 5.2.1.2 is required.
(j) When a V3c or V3s fiberboard box is required for level B (see 5.2.1.1).
(k) When patties cut 90 to the pound are to be packaged in accordance with 5.1.1.3.1.

6.2 Appropriate level of pack (Military agencies). Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.14D/ or DSAR 4145.7, as applicable.

6.3 Intended use of nonmetallic reinforcement of shipping containers. Packing specified in 5.2.1.1 is intended for transfer at sea operations or specific overseas operations.

6.4 Assay for Vitamin A or D. If the assay for vitamin A or D, as applicable, made on the initial production lot complies with the specification requirement, thereafter for a period of four months the supplier need only furnish a Certificate of Compliance (C.O.C.) on subsequent but contiguous contracts to satisfy this requirement. The C.O.C. accepted in lieu of an actual vitamin A assay shall be subject to Government verification on each contract or when special testing is deemed necessary. If the initial production lot does not comply with the vitamin A requirement, testing of subsequent lots shall be continued until two successive lots are in compliance.

6.5 Intended use of alternate shipping containers. Packing specified in 5.2.1.2 shall be used when experience or knowledge of the destination and mode of shipment would indicate that a lower grade of board can be used.

6.6 Intended use and technical information.

6.6.1 Type I. Type I margarine is intended for general table use. Its firmness at serving temperatures, relative ease of spreading when cold, and its desirable eating and cooking characteristics make the product an excellent fat

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

spread. The product is sometimes stored at moderate temperatures without refrigeration. However, like any other table fat spread, margarine keeps its best flavor and consistency when kept cold, covered, and away from strong food odors. When product is stored under refrigeration at 45°F or below (7°C or below) an approximate four month storage life is expected before some evidence of flavor (freshness) deterioration can be expected.

6.6.2 Type II. Type II margarine is superior to type I product for general table use. The product has a more delicate flavor, better spreadability at refrigerator temperatures and is free from waxy meltdown characteristics in the mouth. Refrigeration is required to maintain its superior flavor quality and physical properties. Storage life for type II margarine when kept refrigerated is approximately three to four months. The freshness quality begins to deteriorate thereafter.

6.6.3 Type III. The type III margarine possesses the same sensory qualities of the type II margarine but has a distinctly softer consistency, particularly at chill temperatures of 40°-45°F (4°-7°C) with an excellent, smooth spreading characteristic. Because of its soft and plastic consistency as opposed to the more solid form of types I and II, soft margarine cannot be packaged in print form but is commercially put up in bowls, cups or tubs. These containers are usually made from FDA approved high density polyethylene and the lids are of low density polyethylene composition. One of the commonly recognized nutritive or dietetic advantages of soft margarine is the fact that a larger proportion of the liquid form of vegetable oils, usually high in polyunsaturated fatty acids, may be formulated into the product than is feasible with the printed or solid form of margarine. When held at chill temperatures cited above a shelf life of three to four months may be expected.

6.6.4 Type III. Classes 1, 2 and 3 margarines are designed to posses the same table grade characteristics as for type I except for the substitution of animal fat in part or in its entirety for vegetable oils. A further distinction is made that some animal fat margarines lean more toward the shortening or baking type commonly used in baked goods such as cakes and puff pastries. These specialty margarines are not to be considered as products designed for table grade use.

6.6.5 Type IV. Composition of type IV shall be limited to class 1 product. Except for salt, type IV, class 1 product should equal type II, class 1 product in all quality factors.

6.6.6 Keeping quality tests. It is recommended that keeping quality tests be conducted and corresponding records be maintained for purposes of performance capabilities. At least one sample (1 pound print or its equivalent) should be set aside from each lot and held at approximately 45°F (7°C) for 30 days to assure there is no evidence of oxidized, rancid, or aged flavor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

6.6.7 Definitions and industry terminolgy. The word "patties" as used in this specification is equivalent to "pats" or "chips" as recognized by the margarine industry and normally represent the 72 or 90 count square cut or tri-cut individual servings wrapped by sheets or layers and packed in five pound (2.25 kg) boxes. "Reddies" or "pats" also refer to 72 or 90 count individual servings when placed on individual paperboard "mats" (or "chips", see 5.1.1.3.1) and covered with individual parchments called "hats". These servings are packed in trays of two 1/2 pound (227 gram) trays per pound (0.45 kg).

6.7 Destination inspection (for civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quality only. Unless otherwise specified, such inspections should be made by the receiving facility.

6.8 Award of contracts. Award of contracts for the product specified in this document shall be limited to plants known to maintain the required sanitation conditions of 3.5 (also see 4.2.1).

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

AGR - AMS
VA - DMS
GSA - FSS
HEW - FDA, NIH

Project No. 8945-0049

U.S. GOVERNMENT PRINTING OFFICE : 1977 - 241-233/1125

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 65 cents each.

★U.S. GOVERNMENT PRINTING OFFICE: 1977—241-233:1125

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.